

Attention-Based BiLSTM for Sentiment Classification

by Xiaoqing Yao, Lin Zhou, Linzhuo Zhang

1. Introduction

Sentiment analysis of movie reviews remains a significant challenge in Natural Language Processing, as opinions are often expressed through short phrases, negation, and sparse semantic features. Traditional machine learning methods typically rely on manual feature engineering which often struggle to capture deep semantic relationships and long-range dependencies.

In this project, we implement and evaluate an Attention-based Bidirectional Long Short-Term Memory (AB-BiLSTM) neural network to address these limitations. Based on the architecture proposed for short-text classification, our model integrates a BiLSTM to capture context from both forward and backward directions, paired with an attention mechanism that dynamically weighs the most sentiment words in a sentence.

Our goal was to investigate the synergy between bidirectional LSTM and attention layers. We conducted a comparative study across four architectures: a basic LSTM, a BiLSTM, and their respective attention-enhanced versions. By evaluating these models on the IMDB dataset across different input lengths (50 and 200 tokens), we aim to demonstrate how the AB-BiLSTM framework effectively captures qualitative emotional cues without the need for external linguistic resources or handcrafted features.

1.1. Literature review

Recent research, such as the study by Xianglu Yao, suggests that the integration of BiLSTM and Attention mechanisms can produce synergistic effects in short-text analysis. By simultaneously using bidirectional contextual encoding and dynamic feature weighting (attention mechanisms), this hybrid approach aims to overcome the limitations of simpler recurrent architectures.

2. Methodology

To maintain consistency with our model paper, we implemented and compared four architectures: LSTM, BiLSTM, LSTM + Attention, and BiLSTM + Attention. Each review was

tokenized, mapped to integer indices with a vocabulary, and truncated to a fixed length. The 200-token setting was used as our main experiment because it preserves more review context, while the 50-token setting served as a stress test to evaluate whether the models could still perform well with much less contextual information.

Our main BiLSTM + Attention model first maps input tokens into word embeddings. After embedding dropout, the sequence is passed into a bidirectional LSTM, where the forward and backward hidden states are concatenated. The attention layer then computes importance weights over the hidden states and uses them to form a weighted context vector. Finally, this vector is passed through dropout and a fully connected layer to produce the positive or negative sentiment prediction.

During training, we used Adam with a learning rate of $1e-3$ and selected the best model based on validation accuracy. Early stopping was applied when validation accuracy stopped improving. Our parameter settings mostly followed the original paper. We used the same main structure and hyperparameter choices as the paper, including an embedding size of 50, LSTM hidden size around 100, attention mechanism, and dropout.

2.1. Data

This project utilizes the IMDB 50K movie review dataset (via Lakshmi BN, Kaggle), a benchmark for binary sentiment classification. It contains 50,000 polarized reviews originally collected by Stanford researchers, featuring a balanced 1:1 ratio of positive and negative samples (25,000 each). Because the task is binary classification, each review was assigned a label of either 0 (negative) or 1 (positive). To ensure label reliability, the dataset only includes extreme ratings (≥ 7 or ≤ 4).

3. Challenges

The most significant technical challenge of this project is in the implementation of the BiLSTM-Attention. Specifically, managing multi-dimensional tensor operations proved complex, as it required tracking of shapes across embeddings, LSTM hidden states, and attention results. To convert these dimensions correctly and perform batch matrix multiplications (BMM), we utilized operations such as squeeze and unsqueeze. Any error at this stage would not only trigger runtime errors but also result in incorrect attention weighting.

Furthermore, handling padding tokens within the batching process presented a critical challenge. Since padding tokens have no semantic value, we implemented a masking logic using

masked_fill to set the attention scores of <pad> tokens to negative infinity prior to the Softmax layer. This ensured that irrelevant positions received near-zero weights and did not skew the final sentiment representation.

Beyond implementation, overfitting is the most persistent issue for now. This highlighted the model's tendency to memorize noise within the training set rather than learning generalizable patterns. To resolve this, we integrated regularization techniques, including dropout tuning, weight decay, and early stopping, to enhance the model’s robustness on data.

4. Results

We evaluated our models under two input-length settings: 200 tokens and 50 tokens (as a stress test). The best overall result was achieved by the BiLSTM + Attention model with 200-token inputs, reaching a test accuracy of 88.56%. All results are given below. These results support the main idea of the paper: combining bidirectional recurrence with attention can improve sentiment classification performance.

Model	Test Accuracy(200 token)	Test Accuracy(50 token)
LSTM Baseline	61.02%	78.90%
BiLSTM	87.48%	78.32%
LSTM + Attention	88.03%	78.72%
BiLSTM + Attention	88.56%	79.62%

Figure 1. Test accuracy for four model variants under 200-token and 50-token input settings.

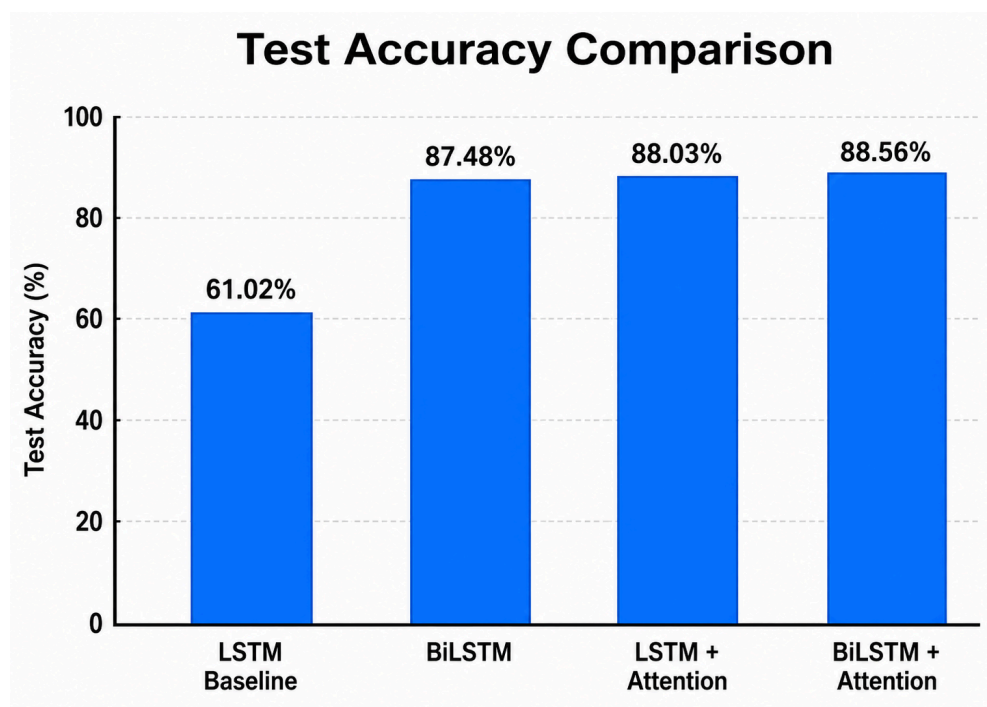


Figure 2. Test accuracy for four model variants under 200-token input setting.

4.1. Interpretation of Results

The 200-token results suggest that both bidirectional context and attention are helpful for sentiment classification. BiLSTM improves performance because it captures context from both directions. Attention improves performance because it allows the model to focus on the most important words instead of treating every hidden state equally.

The 50-token experiment provides an additional stress test. With shorter inputs, BiLSTM + Attention still achieved the highest accuracy at 79.62%, but the gap between models was much smaller. This may mean that the model needs enough context for bidirectional recurrence and attention to be fully useful. If some of the review is truncated, the model may not have access to the words that determine the sentiment.

5. Ethics

This project raises several ethical considerations related to the use of machine learning models for sentiment analysis. First, the IMDB dataset consists of user-generated reviews, which may

reflect biases in demographics, language usage, and cultural perspectives. As a result, the model may learn and reproduce these biases, leading to skewed or unfair sentiment predictions.

Second, although our best model achieves relatively high accuracy, it is not error-free. Misclassification may occur in complex linguistic cases such as sarcasm, negation, or subtle emotional expression. In real-world applications, such errors could lead to incorrect interpretations or decisions if the model is used without proper human oversight.

Third, attention mechanisms may give the impression of interpretability by highlighting important words, but these attention weights do not necessarily correspond to true causal reasoning. Users may over-rely on these signals and overestimate the reliability of the model's explanations.

Finally, while the dataset is publicly available, it contains user-generated content. Ethical use of such data requires awareness of privacy concerns and responsible deployment practices. Additionally, since our model is adapted from a short-text framework and applied to longer reviews, its generalization ability may be limited. Applying the model outside its intended domain without proper validation could lead to misleading outcomes.

6. Reflection and Discussion

Overall, this project successfully achieved its stated objectives by constructing a robust preprocessing pipeline and implementing four distinct model architectures; this enabled us to validate the effectiveness of the Attention mechanism as a pivotal tool in sentiment classification tasks. The final results demonstrate that the BiLSTM + Attention model achieved the highest test accuracy under both experimental settings: reaching 88.56% in the 200-token setting and 79.62% in the 50 token setting. A noteworthy finding is the significant disparity in performance among the models in the 200 token setting: while models incorporating the Attention mechanism maintained robust performance, the standard LSTM (LSTM Baseline) performed quite poorly, yielding an accuracy of only 61.02%. This is likely because, when processing longer sequences, the standard LSTM without any attention mechanism struggles to effectively capture and retain long range dependencies. Our research methodology evolved from high level architectural design to a deep thought about technical issues; specifically, as we faced the overfitting problem, we paid more attention to tune the hyperparameters and methods on it. We introduced "Early Stopping" and "Weight Decay" strategies to further resolve overfitting; however, these strategies proved only marginally effective in resolving the issue, and the models currently still exhibit overfitting. Consequently, given more time, we would continue to tune the relevant parameters to better address the overfitting problem. Additionally, we would conduct a systematic grid search to identify optimal hyperparameters and integrate a wider range of pre-trained word embedding

models. The most significant insight gained from this project is that the synergy between model architectures is important. Although the Attention mechanism consistently identifies sentimental keywords, it is only through the combination of bidirectional contextual information and the attention-weighting mechanism that one can achieve the most robust and superior performance when processing text sequences of varying lengths.

7. Code Repository

Our Github link is: <https://github.com/Lyn9527/CSCI1470FinalProject>

8. References

[1] Xianglu Yao. (2018). Attention-based BiLSTM Neural Networks for Sentiment Classification of Short Texts.